



Exploring Visitors' Behavior and Subscription Trends for a MOOCs Subscription-based service platform (A Case Study- Data365)

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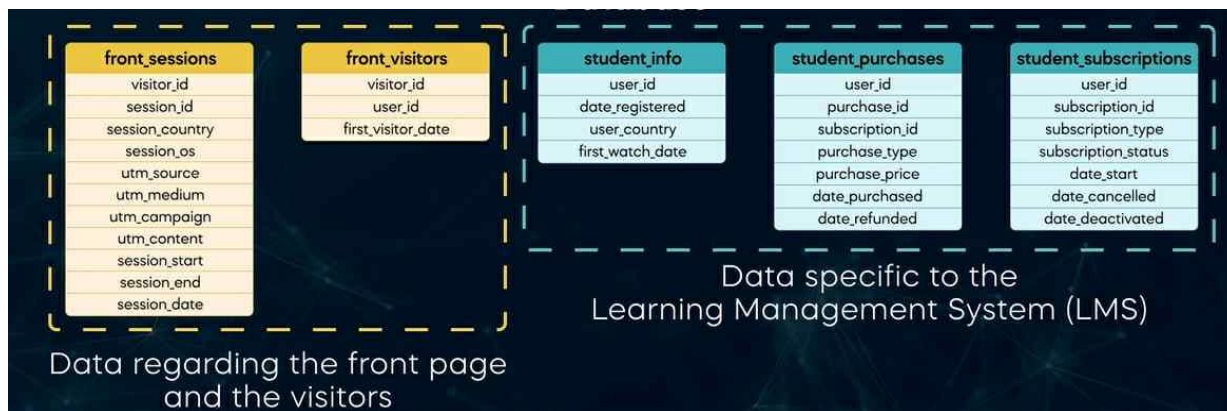
I. Introduction

Subscription-based business models such as Netflix, Canva, and ChatGPT Plus (Wikipedia, 2025) are dominated today's digital space, they attract countless number of visitors, convert them into plan-subscribers, and maintaining/maximizing consumers' retention through meticulous data-driven decisions. Therefore, understanding visitor behaviors/platform data is crucial in today's data-driven, especially in marketer and product manager positions, where they can receive vital data about how visitors/users interact with the platform; thereby creating meaningful story about the platform's data which prompts suitable data strategies for their company (Kourouniotis, 2023). This investigation focuses on a case study of a subscription-based Learning Management System (LMS), Data365; through visualizing and analyzing the trends in new and returning visitors across different acquisition channels, when is the largest drop-off occurs, users' geographical patterns, and which areas/campaigns are effective for attracting new paid-users, gaining deeper insights on how customers/potential customers interact with Data365 platform.

Personally, my motivation lies in learning how data visualization and analytics can be transformed into actionable monetizing business strategies, especially within the education technology sector, where insights into learner behavior can directly inform growth strategies, enhance student engagement, and improve revenue income. As competition intensifies, the platform must leverage data-driven insights to create an competitive advantage, sustaining customers loyalty, and harmoniously attracting new customers in a timely manner. Three main questions that this report will focus on are:

1. What trends can be identified in new and returning visitors across different acquisition channels over time?
2. How does the distribution of mobile vs. desktop usage vary by country, are there any discernible geographical patterns?
3. What proportion of visitors convert to paid subscribers, and at which stages of the funnel do the largest drop-offs occur?

II. Data Wrangling and Checking



The temporal data is originated from Data365, collected from 1st April 2020 until 31st December 2022, comprising 5 tabular data sources, including front_sessions, front_visitors, student_info, student_purchases, student_subscriptions. front_sessions and front_visitors collected when a user interacted with the platform Webpages, whereas student_info, student_purchases, and student_subscriptions collected when a user interacted with the platform's product, which is a Learning Management System (LMS). The requirement for a visitor to become a user is when they signed up for an account in the platform (even a free-trial); the purchase_type and subscription_type are separate attributes with four distinct values: 0 for monthly, 1 for quarterly, 2 for annually, and 3 for lifetime. The purchase_type indicates when the user first starts paying for the subscription, while the subscription_type details the user's ongoing journey and the current status of that subscription, which can change over time.

Since the data is extensive, documenting in an SQL files, where front_sessions (728658 records), front_visitors (669526 records), student_info (263885 records), student_purchases (19131 records), student_subscriptions (9655 records), the URL will show a glimpse of the dataset: https://github.com/HuyJonesNguyen/Subscription_based_model-visualization

Data Checking

Since the data is SQL injected, the type of attributes are coherence and robust, with only minor issues related to missing data and data noises (be could be addressed in Data Wrangling procedure). The procedure in data wrangling process is name after the Figure ID in III. Data Explorations.

Data Wrangling Procedure

Procedure II.1.1:

The data extracted from SQL by joining two tables, `student_info`, provides the number of user registration and `student_purchases`, provides users' transaction history, where free users are not having a purchased date in `student_purchase` and the `purchase_type` is not in range (0,1,2,3) while the time difference between (registered and buying the plan) must larger than 30 mins (otherwise it could be considered as someone who already interacted with the LMS product buying it). From former process, number of free users and paid users are created corresponding to the `date_registered`; the conversion rate is calculated by dividing the paid users (converted) over free users.

Procedure II.1.2, II.1.2.2, II.1.3:

Since the `front_sessions` dataset is extensive and cannot be exported fully via csv file, SQL code was employed for dimensionality reduction and targeted retrieval. Noticeably, the date format in the database is recorded as Datetime format; therefore in this procedure II.2, the `session_date` is casted to corresponding Date value using SQL date functions. Furthermore, during data cleansing, there are 46 identified record outliers, such as "ebjaybbebcyf", "fbdvby", "ILzJA", etc. as an `utm_medium` and "aDqCFYfUUeSb", "ebgb-fdvfadf-vagfeivfj-dhffgvgbaf-cea", "aDqCFYfU", etc. as an `utm_source` (likely bot traffic); these values did not align with any known traffic acquisition categories (e.g. "cpc", "Direct Search", or "referral"); therefore, to avoid data corruption, such records have been removed from the wrangling dataset. Besides, there are some occurrences of equivalent term in `utm_medium`, including "sm", "social", "Social Media", which eventually convert to same category "Social Media". Finally, using count function to calculate the number of visitors per channel; aside from that, the `front_session` has been collected from 07/06/2022-20/09/2022 only, which did not capture the campaigns, medium on the month of the peaks.

Procedure II.1.4:

At First, extract `user_id`, `subscription_id`, `purchase_type`, `subscription_status`, `subscription_start`, `subscription_end` from `student_subscription` table. Then, a python document have been written to classify users into 4 different categories, including "Base" (the number of current user), "New" (the number of new paid-subscribers), "Churned" (the number of paid-subscribers have cancelled the subscription), and "Reactivated" (the number of returning customer from subscription-cancellation) (Pulev, 2023) . If users already active at the beginning of the period (Base), then flags those whose first subscription begins during time the data collected (31st Oct - 31st Dec 2022) (New), users whose subscriptions end within the period are marked as Churned, while those with subsequent (non-first) subscriptions restarting in the specified time are flagged as Reactivated (same person can be in in two or more categories) (Pulev, 2023).

Procedure II.2.1, II.2.2, II.2.3

By joining three tables `front_sessions`, `front_visitors`, and `student_info`, to extract the geolocation reach of 365 visitors and paid users, also classify `session_os` (from `front_sessions`) into mobile and PC count, by using `startswith` (LIKE clause with string prefixes, such as "iOS%", "Android%" are mobile and "Win%" "Linux%", etc. are PC). The generated dataset is then exported and then analysed in Tableau using geospatial visualisations (choropleth and proportional symbol maps) to show regional traffic patterns and device usage distributions.

Procedure II.3.1, II.3.2, II.3.3:

The datasets for Sankey diagram was obtained through extracting through classifying user into groups of (Watch vs Not watch, Registered vs Not registered and `purchase_type`) through grouping statement in SQL. The temporal alignment of data sources was an important issue in this process: while LMS data span a longer period of time, front-end session records are only available from 07/06/2022 to 20/09/2022. Therefore, while merging and extracting numbers from these sources, the analysis was limited to records dated after July 6, 2022, to ensure consistency and validity in the flow representation.

III. Data Explorations

1. New and returning visitors across different acquisition channels

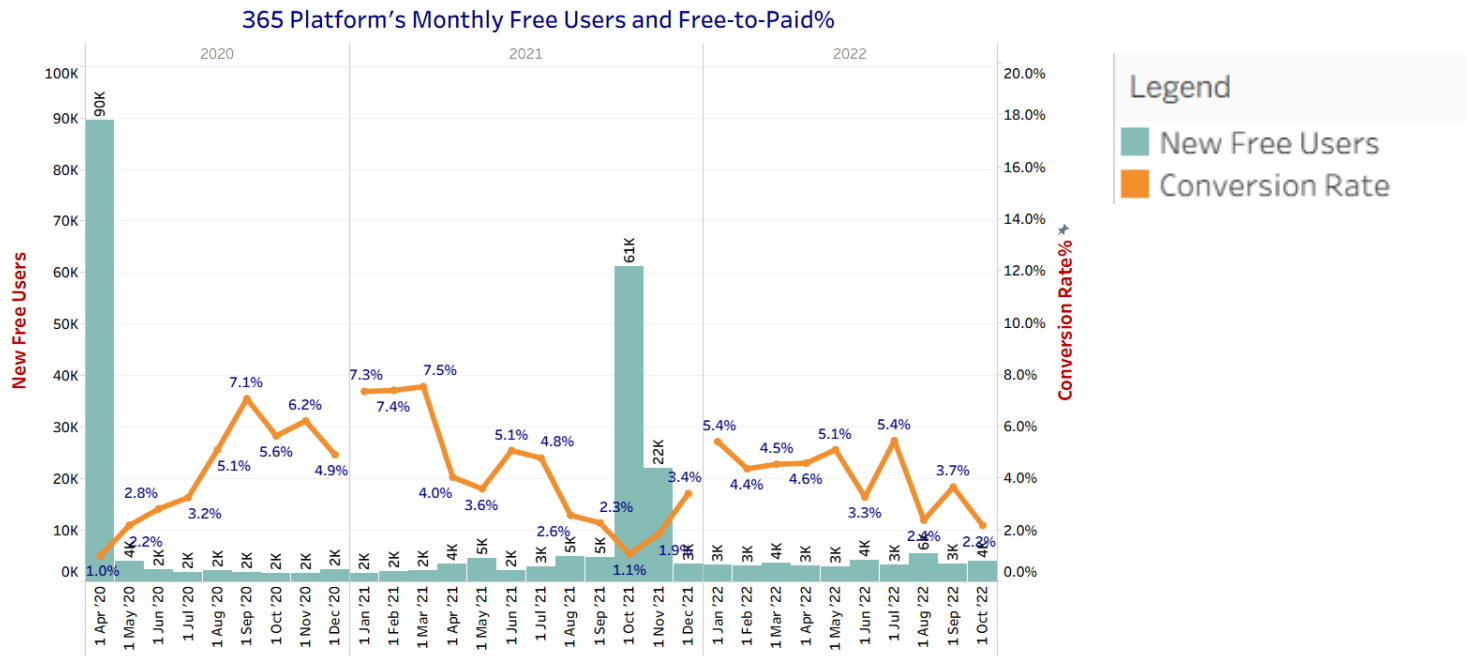


Figure III.1.1: 365 Platform's Monthly Free Users and Free-to-Paid% (from 1st April 2020 to 1st Oct 2022)
Source: Huy Nguyen, 2025

Figure III.1.1 illustrates 365 Platform's monthly free user acquisition and free-to-paid conversion rate between April 2020 and October 2022, revealing two major surges in free users visiting 365 platform, April 2020 (approximately 90 thousand users) and October 2021 (approximately 61 thousand users). However, both peak coincided with considerable low conversion rates (1% and 1.1%, respectively), suggesting that these increases were driven by external marketing campaigns, promotions, or even pandemic-related demand, resulting in high-volume but low-intent traffic. Additional to this, Panetta (2021) stated that 2020 and 2021's are accumulating the hype for emerging technologies related to big-data, could possibly reasoning the two peaks coincided. Notably, in the months following each peaks, the conversion rate rebounded upward, reflecting a more selective pool of users progressing through the subscription funnel; such rebound is consistent with patterns typically observed in subscription-based MOOCs learning management systems, where initial surges attract a large volume of exploratory users, followed by a natural filtering process in which committed learners convert at higher rates (Figure III.1.1).

Aside from these oddities, new free user acquisition remained stable at 2K-5K monthly, but conversion efficiency increased, peaking at 7.5% in March 2021. However, by latter half of 2021, after the global peak conversion rate, the users' conversion rate gradually declined to 2.3% by the end of September, but increasing in more free visitors, this might be due to smaller marketing campaigns such as (paid Media/ Influencers, or more small marketing campaigns, etc.), Nevertheless, by 2022, both acquisition and conversion metrics worsened, with free users averaging approximately 3-4K per month (with an exception 6K on the August 2022) and conversion rates plummeting to 2-5%, reaching as low as 2.2% in October 2022 (Figure III.1.1); this trend raises concerns about the platform's growth viability, as it is not considerably expanding its free user base or maintaining its conversion efficiency. The data reveals a the juxtaposition of rising acquisition with persistently low conversion (under 8.4%, normally for LMS (Unbounce, 2024)), large-scale campaigns increase sign-ups but decrease conversion rates, whereas smaller, organic growth periods result in higher monetization (aligns with Panetta (2021) research for the hype cycle).

365's Visitors Distribution via Marketing Channels

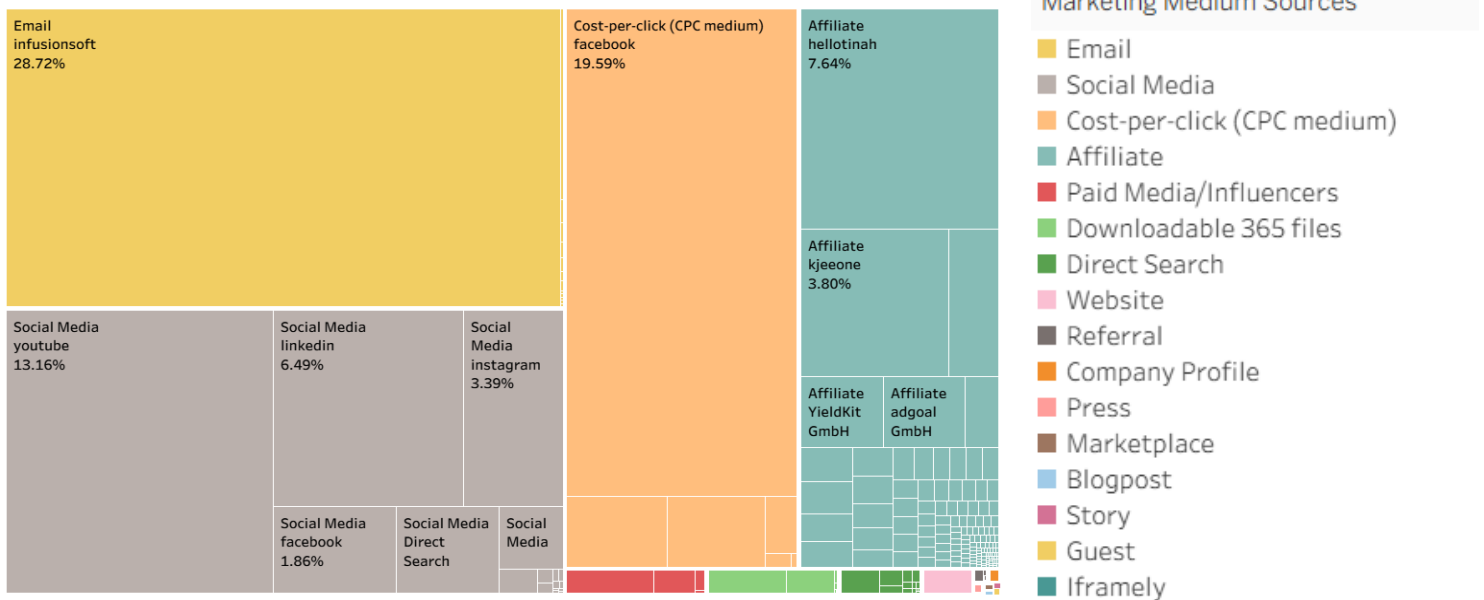


Figure III.1.2: Platform's Marketing Sources (excluding Google Direct Search) (from 07/06/2022 to 20/09/2022)
Source: (Huy Nguyen, 2025)

Figure III.1.2 visualises 365 Platform's visitor distribution across Marketing Medium Sources, grouped by acquisition categories such as Email, Social Media, Cost-per-click (CPC), Affiliate, Paid Media/Influencers, Direct Search, Referral, and Website-based traffic; this treemap uses areas to encode the contribution of visitors visiting and color hue (qualitative) for utm_medium sources. It is noteworthy that Google Direct Search dominates with 86.13% of visitors traffic, but since it does not capture platforms or campaigns that lead users to perform direct searches, the figures had been excluded for scrutinizing other marketing sources. Overall, the data demonstrates that "Email", "Social Media", "CPC medium", and "Affiliate", accounting for 28.83%, 27.42%, 22.4%, and 19.35%, respectively. Precisely, via "Email", Infusionsoft contributes 99.7% of the whole "Email" traffic, showing that there is a primary partnership between two companies (365 and Infusionsoft), or a long-term contract to train their employees. Notwithstanding, Facebook dominates "CPC medium" with 19.59%, equivalent to 87.46% of "CPC medium" channel, and 15.8 times higher than the second "CPC" channel which is "LinkedIn", accounting for only 1.24% share in total. However, despite the domination on CPC marketing source, 365 Facebook pages does not perform as well, falling short at 4th place in "Social Media" medium, staying behind Youtube, LinkedIn, and Instagram, equivalent to only 14% compared to this figure at Youtube (Figure III.1.2).

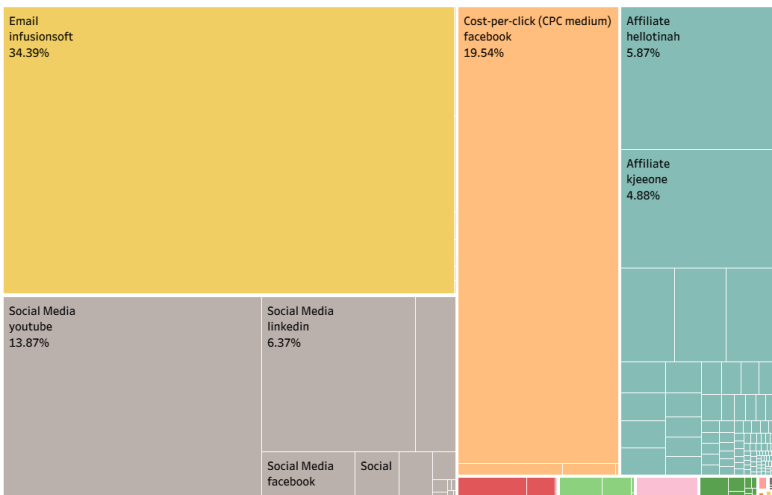


Figure III.1.2.2: Platform's Marketing Sources (excluding Google Direct Search) (August 2022, with same legend with Figure III.2)

Smaller medium sources, including Paid Media/Influencers, Blogposts, Guest, Press releases, and Marketplaces, represent long-tail channels with lower but potentially higher-quality for niche visitors, suggesting untapped opportunities for brand building and sustainable traffic growth. Finally, the distribution of marketing sources' traffic reveals a concentrated acquisition strategy around outbound channels (Email and CPC), while the remainder is fragmented across self-branding and affiliate-driven sources (Figure III.1.2).

August 2022, stood out with approximately 6K in the number of visitors within the available data dates (Figure III.1.1). But, compared with Figure III.1.2, there's no evidence for extensive changes in Figure III.1.2.2 Marketing medium sources, except Infusion soft take a larger stake overall, suggesting a trajectory for the platform to shift target customers to tech companies.

Number of visitors and return% for top 9 marketing medium sources

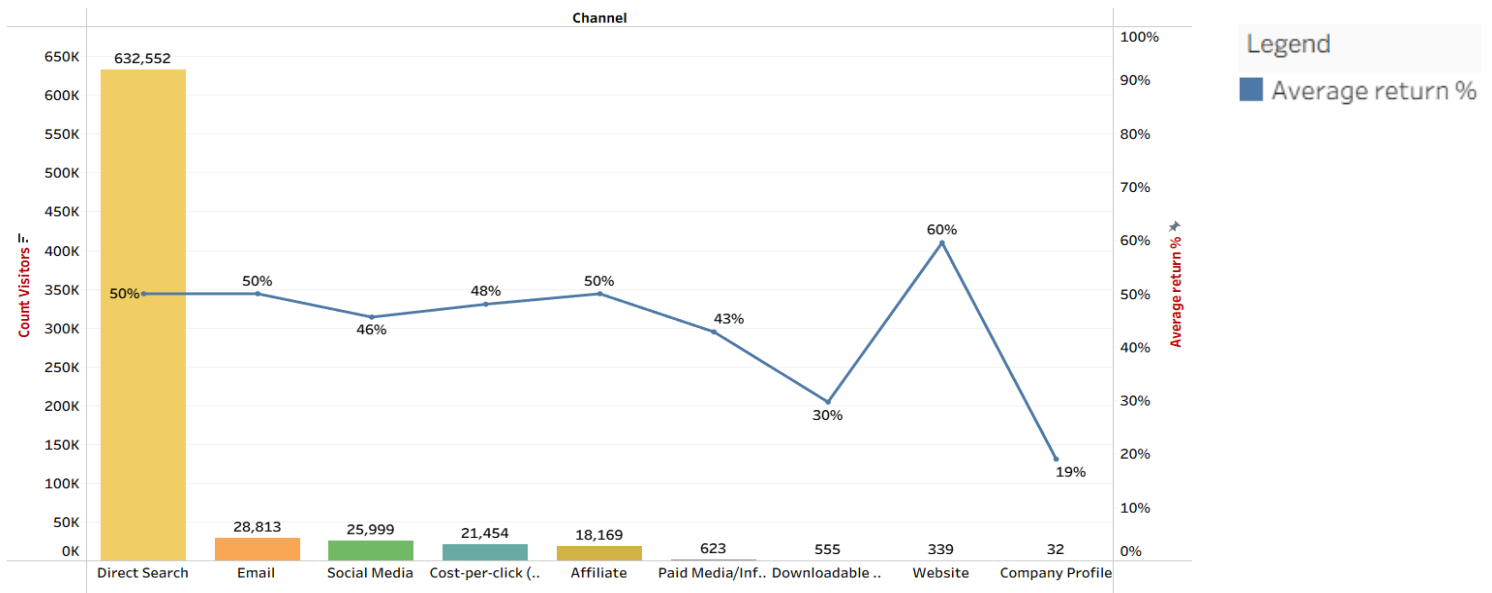


Figure III.1.3: The number of visitors and average return rate (%) for the top 9 marketing medium sources
Source: Huy Nguyen, 2025

Figure III.1.3 illustrates the number of visitors and their average return rate (%) across the top nine marketing medium sources (each medium must have at least 30 visitors), utilizing dual-axis format (combined view of both acquisition scale and engagement efficiency). Direct Search overwhelmingly dominates traffic acquisition, with 632,552 visitors, far exceeding all other channels; nevertheless, its return rate averaged around 50%, reflecting its nature as a brand-driven or the platform's content driven. By contrast, Email (28,813 visitors) and Social Media (25,999 visitors) represent the largest campaign-driven sources, with return rates of 50% and 46% respectively. Following that, Cost-per-click (21,454 visitors, 48% return) and Affiliates (18,169 visitors, 50% return) demonstrates the paid and partner channels in broadening reach, albeit with slightly lower engagement efficiency than "Direct Search" pathways. Interestingly, Website referrals (339 visitors) show disproportionately high quality, achieving the highest return rate of 60%, while Company Profile traffic (32 visitors) performs poorly with the lowest return rate of 19%. From a strategic perspective, the results suggest that 365 Platform should leverage Direct Search and Email for volume, but simultaneously invest in strengthening high-intent, organic channels such as Website referrals to maximise retention.

Students' subscriptions status in 2022 (compared to 2021)

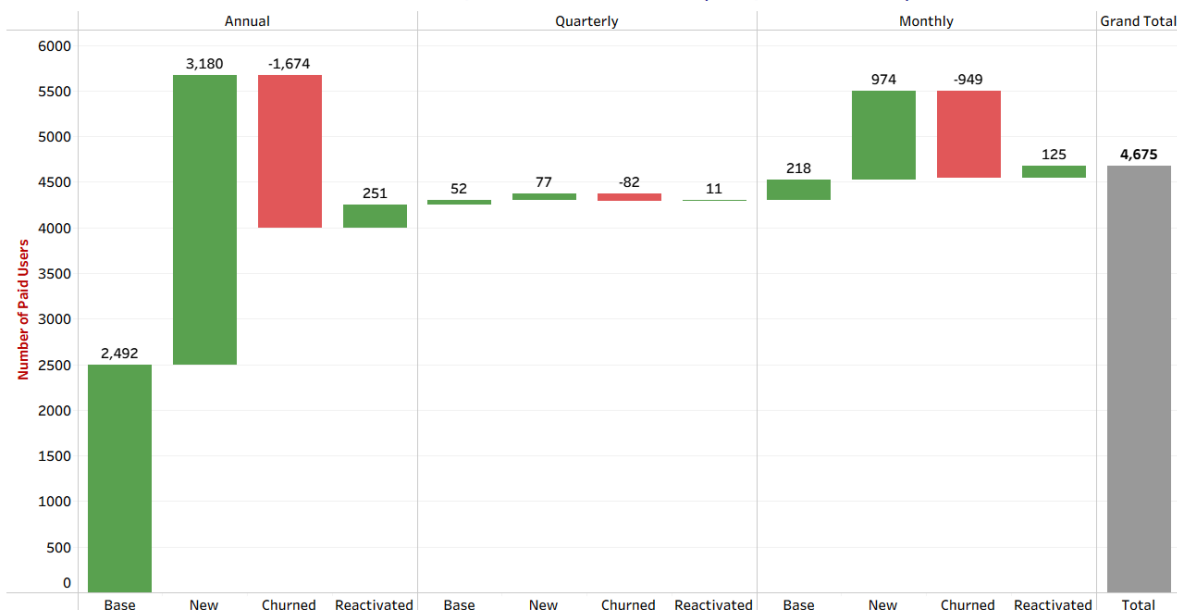


Figure III.1.4: Students' subscriptions status/plans in 2022 (compared to 2021)
Source: Huy Nguyen

Figure III.1.4 visualises the status of students’ subscriptions in 2022 compared with 2021 (on plan-types, Annual, Quarterly, and Monthly), encoding net changes in the number of paid users across four stages: existing subscribers from 2021 (Base), newly acquired subscribers (New), those who cancelled or did not renew (Churned), and those who returned after cancellation (Reactivated). Overall, in comparison with 2021, the current number of paid-subscribers increased 187.6% in the number of paid-subscribers, from 2492 paid-users upto 4675 paid-users (excluding users with lifetime access). The plan with highest volume is Annual Plan, witnessing 3180 new paid subscribers and 251 reactivated users, whilst the churned rate is approximate 29.5% ($\text{Churned}/(\text{Base} + \text{New})$), the reactivation might occurs from other plans). By contrast, the Quarterly plan showed the smallest volume amongst all, just a modest uptake (77 new, 11 reactivated, and 82 churned), indicating a lack of user interested in this plan, indicating the pricing strategy is not ideal for this plan. Besides, the Monthly plan exhibits instability, with 974 new sign-ups virtually balanced out by 949 churned users; 125 reactivations partially offset losses, but the significant turnover indicates the vulnerability of short-term subscriptions (this might be the customers are not further committed to 365 LMS, the platform targeted at wrong audiences, or even the hype of big-data is fleeting).

2. Discernible geographical reach of 365 platform visitors/users

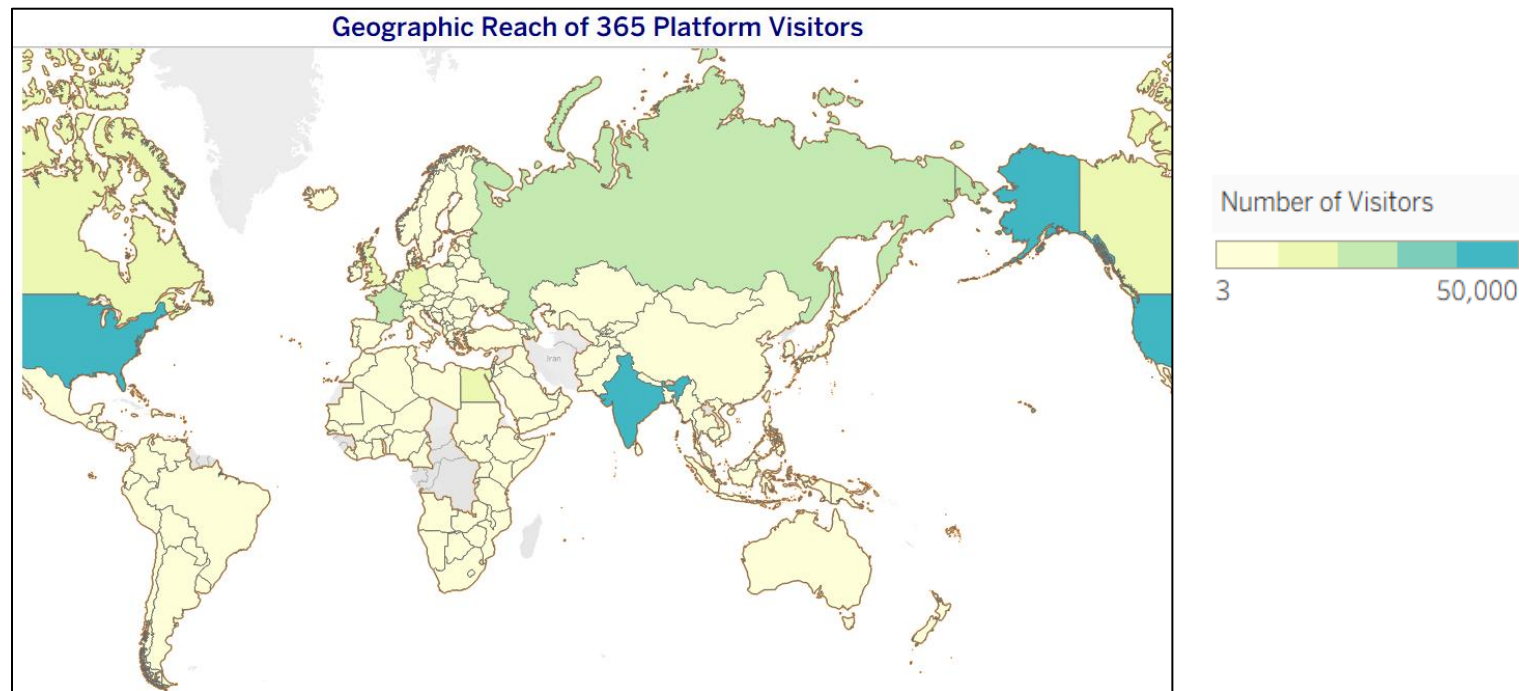


Figure III.2.1: Geographic Reach of 365 Platform Visitors
Source: Huy Nguyen, 2025

This choropleth map (Figure III.2.1) illustrates an intuitive overview of the platform’s visitor footprints, highlighting regions of strong visibility as well as areas with minimal exposure to the platform, with colour sequential representing the relative volume of traffic by country. Figure III.2.1 shows that the United States, India, and Russia are amongst the country with the most 365 platform’s visitors, accounting for 117,935 visitors, 76,321 visitors, and 23,906 visitors, respectively; these areas reflect both high levels of data-driven solution adoption and greater demand for online learning resources. Moderate engagement is observed in Western Europe, Canada, and East Asia, suggesting established but less concentrated adoption. In contrast, large parts of Africa, South America, and Central Asia display light shading or no color, underscoring potential targeting regions where platform has limited visibility. Strategically, the map shows strengths and growth prospects, where the concentration of visitors in the U.S., India, and Russia confirms the platform’s ability to penetrate large, digitally active populations, while the limited exposure in Africa, South America, and Central Asia. Additionally, 365 platform could localized content and offering affordable subscription models to regions such as Africa and Latin America, where online education demand is growing.

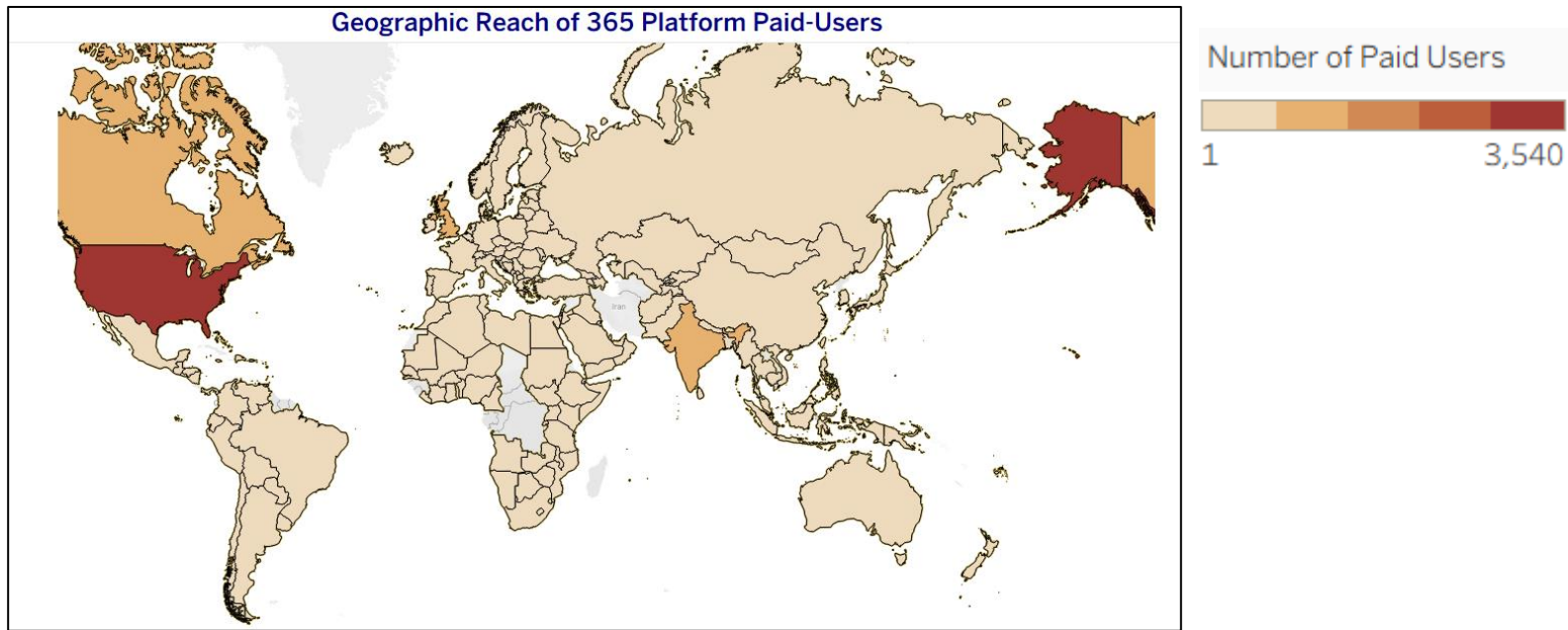


Figure III.2.2: Geographic Reach of 365 Platform Paid-Users

Source: Huy Nguyen, 2025

The choropleth map (Figure III.2.2) illustrates the number of subscribers per country through colour intensity, highlighting the regions with strong engagement as well as areas with minimal exposure. Figure III.2.2 shows that the US is the most significant market, with the darkest intensity indicating the highest volume of 3540 paid-subscribers. Following that, India has 860 paid-users, Great Britain has 758 and Canada has 723 paid-users, showing early stage of data LMS adoption around North America, Canada, India and numerous Western European countries. Conversely, much of Africa, Central Asia, and parts of South America show lighter shading or absence of data, indicating limited adoption or lack of visibility in these regions. Notably, Australia has 380 paid subscribers, which contrasts with its relatively low-to-moderate visitor exposure in Figure III.2.1, suggesting higher conversion efficiency compared to its traffic base. Although the absolute number of paid subscribers worldwide remains modest, the distribution demonstrates that Data365 has successfully established a global footprint, with evidence of market penetration spanning multiple continents.

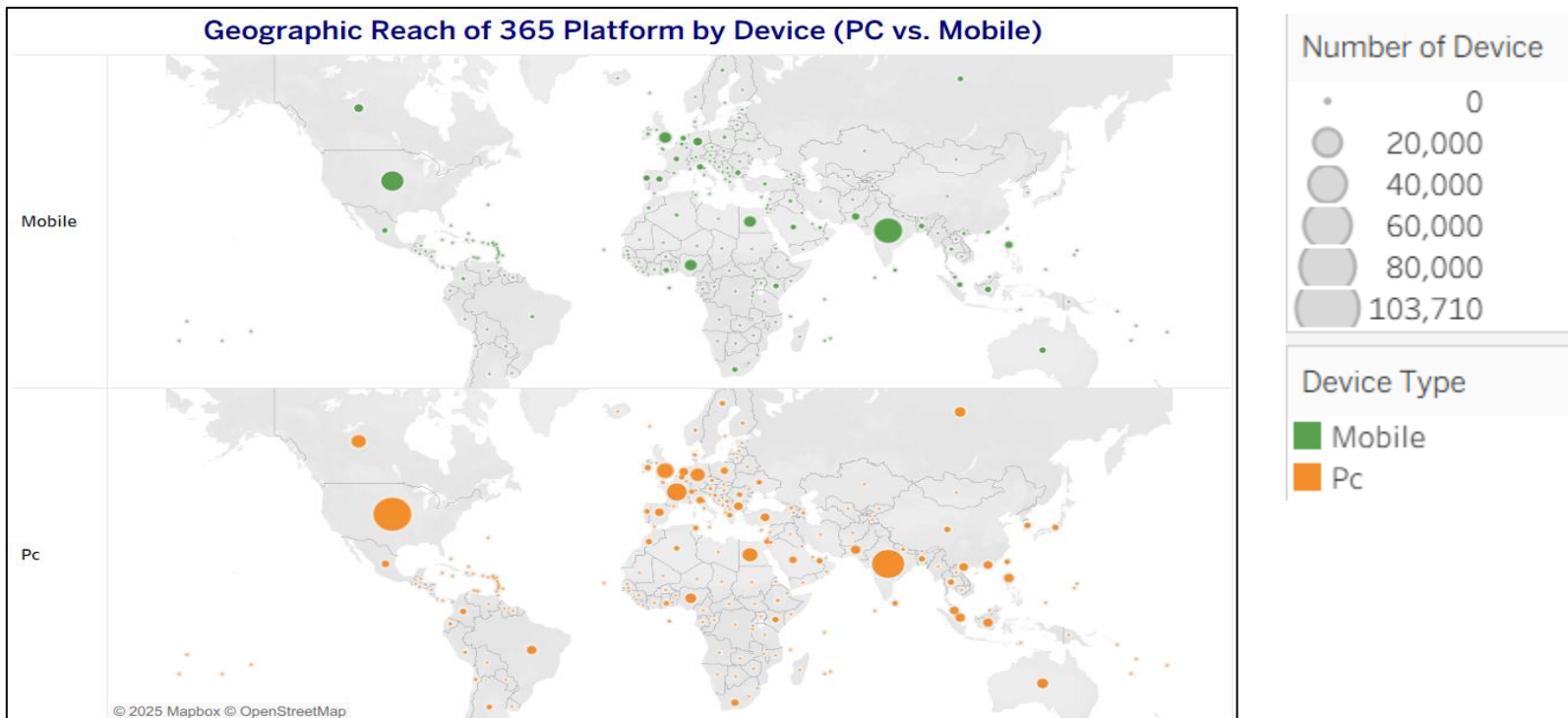


Figure III.2.3: Geographic Reach of 365 Platform by Device (PC vs. Mobile)

Source: (Huy Nguyen, 2025)

The proportional symbols map (Figure III.2.3) depicts the geographic reach of 365 Platform visitors by device type, where bubble size encodes the number of visitors (from none to 103,710 devices). Figure III.2.3 indicate that PC dominates globally, with especially large visitor volumes from the United States, India, and Western Europe, meaning the platform is primarily designs for PC learners. Similar to Figure III.3.1 and Figure III.3.2 findings, the US, India, Canada, , and parts of Western Europe emerge as the strongest market. However, it’s surprising that France and Denmark have comparative PC visiting with Great Britain when Figure III.3.1 shows low number of users, meaning the learners in these regions are more dedicated toward 365 content, suggesting more marketing campaign on these areas. Similarly, Egypt, Nigeria, and several Asia countries exhibits the same trends with highly dedicated learners relative to their overall visitor base, suggesting price sensitivity issues in these regions.

3. Visitors funnel flow

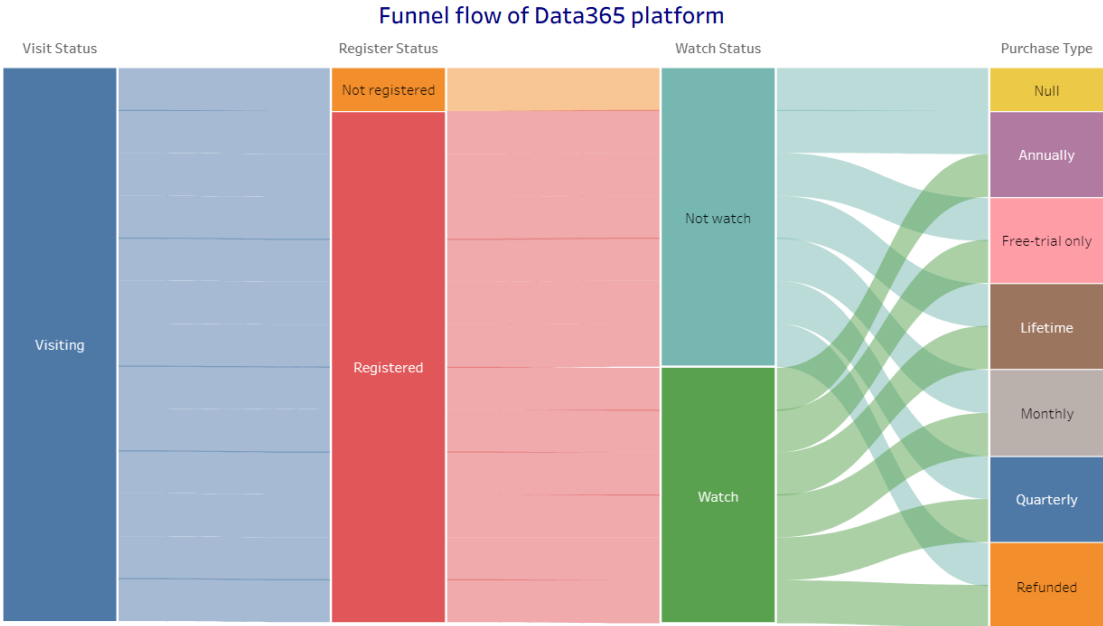


Figure III.3.1: Funnel flow of Data365 platform (only display the flow)
Source: Huy Nguyen, 2025

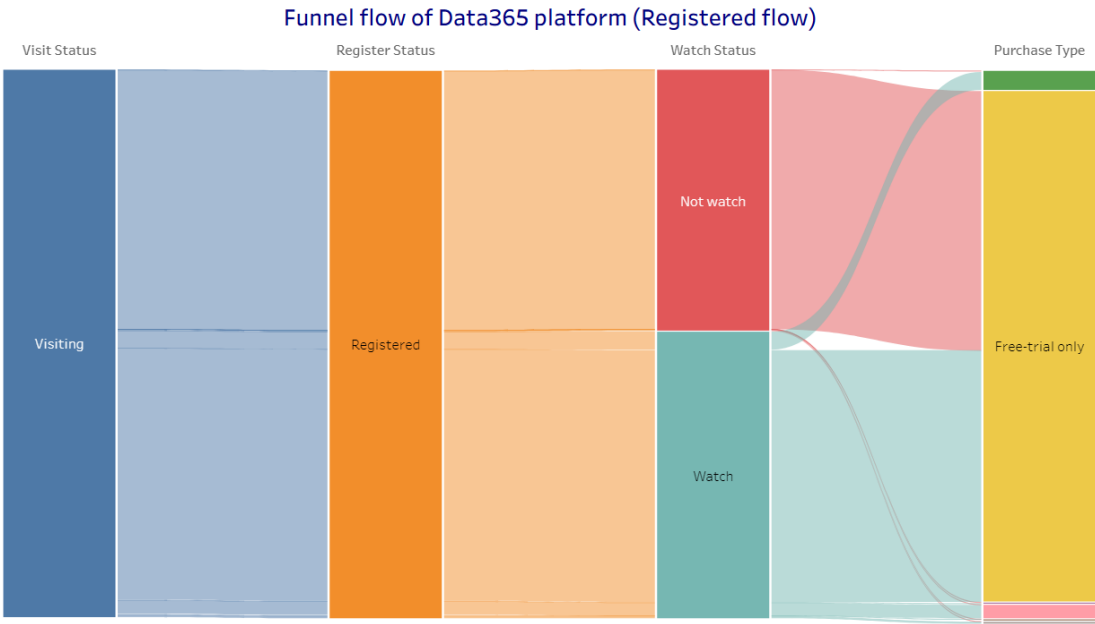


Figure III.3.2: Funnel flow of Data365 platform (Registered concentrated)
Source: Huy Nguyen, 2025

The Sankey diagram (Figure III.3.1) explains the funnel flow of the Data365 platform, tracking user through registration transition into purchase outcomes, following four sequential stages from Visit Status → Register Status → Watch Status → Purchase Type. From visitors, the user must register an account in order to interact with the product in the platform, then

the user would have the option to interact and experiment with 365 LMS product (mainly online course videos), subsequently decided to subscribe or ending the free-trial session. Figure III.3.2 captures the visitors’ registered flow from when a visitor registered an account to become 365’s user to the plan outcomes; it’s noteworthy that most of user (approximately 93% in total) remain concentrated in the Free-trial only base, underscoring limited conversion into paid plans. Moreover, prior interaction with platform content (Watch vs Not Watch) appears to have little influence on whether user continue beyond the free-trial, suggesting that the current onboarding and trial experience does not sufficiently encourage long-term subscription uptake. Notwithstanding, outside of such “Registered flow”, around 97.06% of visitor have no interaction with the platform in anyway.

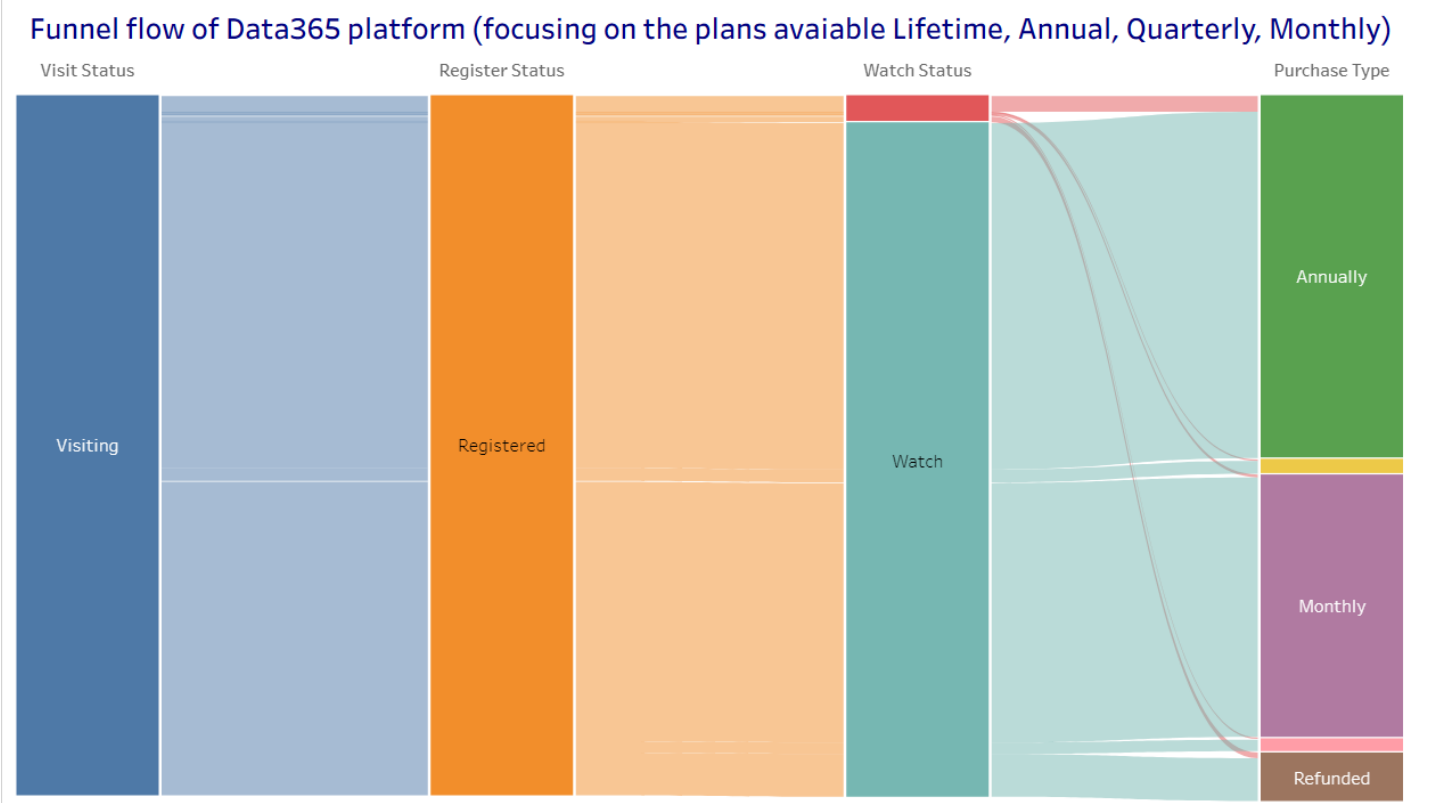


Figure III.3.3: Funnel Flow of Data365 platform (focusing on (Lifetime, Annual, Quarterly, Monthly)

Source: Huy Nguyen, 2025

Figure III.3.3 focuses on the platform’s revenue generating plans (Lifetime, Annual, Quarterly, Monthly); majority of users interested in Annually and Monthly plans, accounting for 51.77% (49.46% from “Watch” and 2.3% from “Not watch”) and 37.4%(37.02% from “Watch” and 0.38% from “Not watch”), respectively. Despite the large stake of the plans in the flow, the minor “Not watch” rate could be potentially due to the interactions that user have had with the product from another account. Besides, it is noticeable that refunded requestion is more favorable than Lifetime and Quarterly plan, accumulated for 6.91%, suggesting a critical need for trial-to-paid conversion interventions and post-purchase engagement strategies to reduce refunds.

IV. Conclusion

In summary, Data365’s platform reveals that while acquisition volumes occasionally surged and peaked (at April 2020 and October 2021), are accompanied by very low conversion rates, reflecting high-volume but low-intent traffic, ranging from 2-7.5% (which is less than average educational MOOCs conversion rate, ~8.4% (Unbounce, 20224)). Marketing channels are highly concentrated in Direct Search (over 86%), with Email, Social Media, CPC, and Affiliates contributing additional traffic but showing mixed engagement efficiency, while smaller channels such as Website referrals generate fewer visitors but higher retention, highlighting opportunities for organic, high-quality acquisition (Figure III.1.3). The three maps illustrates strong markets in the US, India, and Western Europe contrast with limited exposure in Africa and Latin America, though certain regions, including France, Denmark, Egypt, and Nigeria show disproportionately strong engagement potential. Most critically, funnel analysis shows that 97% of visitors never interact meaningfully with the platform, and 93%

of registrants remain in free-trial without converting to paid. Taking everything into consideration, Data365 should prioritize enhancing trial-to-paid conversion by perhaps attracting the right customers via personal-driven marketing sources, and tempted to diversify user's acquisition towards more engaged, high-intent individuals in growing countries.

V. Reflection

There's a story of Data365's platform performance in my chain of figures, beginning with visitor acquisition trends and channel composition, then shifts to subscription dynamics and geographic patterns, followed by Sankey diagrams that visualise the end-to-end funnel (what stage is the most drop-off occurs). This nature of my data relies heavily on visualization instead of statistical analysis, as the majority of data fallen into the drop-offs stages where users are not interest in the content or going for free-trial only (customer behavior datasets would be suitable for complex statistical analysis); therefore, this report is extensively using Tableau due to its appealing/intuitive visualization nature.

Figure III.1.1 and Figure III.1.3, I chose to dual-axis (bar and line) chart is due to the large volume of visitor figures, typically low conversion rates/ moderate return %, and the suitable representation through time-series data, redeeming myself to opt for such choices (visually appealing through high contrast color, with the annotation texts). On Figure III.1.2 and Figure III.1.2.2, I chose to use treemap diagram (distinguish with qualitative colors via utm_medium sources) due to the comparable proportional distribution nature of visitors acquisition channels necessitates to be presented; even though I excluded the enormous portion of "Direct Search" in this section (>85%, and add no additional information on marketing campaign), it is one of the best fit. Figure III.1.4, I utilized waterfall chart to present the new, churned, reactivated users across different plan (except for lifetime plan as they will never churned), distinguished by different color as positive growth displays as green and negative as red (common visualization). Besides, the choropleth maps (Figure III.2.1 and Figure III.2.2) and the proportional symbol map (Figure III.2.3) were chosen to visualize geographical distribution of platform usage and subscription behaviour, encoding intensity by colour saturations (Figure III.2.1 and Figure III.2.2) and color hue (Figure III.2.3), allowing for immediate comparison of relative adoption levels across countries. Precisely, choropleth maps (color saturation, blue-green and orange-gold sequential) effectively emphasises regions of high engagement and areas with limited exposure immediately regarding to its geolocation (that's the reason why I chose choropleth maps over other options such as bar chart and packed bubble). In addition, the dual approach in the proportional symbol maps allow a more precise sense of magnitude while distinguishing device usage (PC vs. Mobile through colour hue) when they have the same legend/measurement, balancing the need for both regional comparisons and volume-based clarity. Finally, for Figures III.3.1 to III.3.3, Sankey diagrams were adopted to visualise the user journey through sequential funnel stages, from visiting and registration to watching content and eventually subscribing. Sankey is particularly suitable here as it highlights both the proportional magnitude of flows and the attrition occurring at each funnel check-point, visualizing the pronounced drop-offs at Register → Paid and Watch → Register.

If I were to redo this assignment again (this is my intuitive thoughts based on what I have known so far), I would try to integrate cloud-based SQL (AWS or Microsoft Azure) directly into Tableau, neglecting the drop-off data along the process from SQL → CSV export → Tableau (cause SQL sever could not export extensive data via csv fully) and allow for more dynamic dashboards (allowing higher reproducibility and scalability). Secondly, SQL skills require for accurate data extraction are undoubtedly valuable, improving the accuracy of the metrics, the data visualization diagrams, and also the statistical analysis (if possible), reducing extensive manual wrangling in Tableau. Thirdly, the importance of data collection and storing is fundamental, following GIGO (Garbage in garbage out) principles, a more robust and dynamic dataset would yield more opportunities for higher/profound visualizations, from here, gaps in dataset/ would be easier to be identified, allowing quicker respond to patch the narrative. Building on this premise, the front_sessions dataset only contains visitor data from 07/06/2022 to 20/09/2022 which could not capture two of the peaks free user, giving us information on the effective marketing utm_source, utm_medium or utm_campaign; or the method of storing attributes in student_purchases and student_subscriptions is not optimal (as date_refunded should be in student_subscriptions instead of student_purchases) for easier data extraction process.

Notwithstanding, due to limited technical skills, I could not present proportion figures onto Sankey diagram on Tableau, enhancing users' interpretability, which I could have done if I understand Tableau a little bit better. Furthermore, I could have had a comparison analysis with other company if I could obtain the desired datasets. Lastly, I regretted that I could have delved in the distribution of visitors via marketing sources on the top paid (or conversion rate) countries, this would yield incredible insights for 365 platform stakeholders; but, due to the stacked 10 pages limit, I couldn't proceed with more visualization figures.

VI. Bibliography

1. Panetta, K. (2021) *5 trends drive the gartner hype cycle for emerging technologies, 2020*, Gartner. Available at: <https://www.gartner.com/smarterwithgartner/5-trends-drive-the-gartner-hype-cycle-for-emerging-technologies-2020> (Accessed: 06 September 2025).
2. Unbounce (2024) *Education and Higher Ed Industry Conversion Rate Trends, statistics and benchmarks*, Unbounce. Available at: <https://unbounce.com/conversion-benchmark-report/education-conversion-rate/> (Accessed: 07 September 2025).
3. Pulev, N. (2023) 'Data365', *Classifying users according to their status (tag_created)*.
4. Kourouniotis, A. (2023) *Seeing data: Using visualization to reveal insights and make decisions | educause review*, EDUCAUSE. Available at: <https://er.educause.edu/articles/2023/2/seeing-data-using-visualization-to-reveal-insights-and-make-decisions> (Accessed: 09 August 2025).
5. Wikipedia (ed.) (2025) *Subscription business model*, Wikipedia. Available at: https://en.wikipedia.org/wiki/Subscription_business_model (Accessed: 10 August 2025).